

Overlapped vs. Non-overlapped Population Comparison

Monte Carlo simulation results (bootstrap coverage)

Replications R = 500 Sample size N = 2000 Bootstrap resamples B = 500 True tau = -1.7500
Propensities: true DGM logistics (fixed). Blips gamma1, gamma2: centered g-estimation, refit per resample.
Runtime: 14.4s on 10 parallel workers. Generated 2026-06-22 15:56.

Target contrast $E(Y^{\{0,1,0\}} - Y^{\{1,1,0\}})$

estimator	mean	bias	emp.Var	cover95	ci.width	RMSE
Proposed	-1.7619	-0.0119	0.0288	0.948	0.6751	0.1699
Partial IPW	-1.7588	-0.0088	0.0347	0.956	0.7353	0.1863
Fully IPW	-1.7503	-0.0003	0.2467	0.944	1.8736	0.4962
Oracle	-1.7578	-0.0078	0.0276	0.948	0.6613	0.1660

emp.Var = variance of the R point estimates (ddof=1). ci.width = mean 95% CI width.
cover95 = fraction of reps whose PERCENTILE BOOTSTRAP 95% CI [q2.5, q97.5] covers the truth
(B resamples per rep, blips refit each resample) -- nonparametric bootstrap, not a Wald CI.

Nuisance reliance:

- Proposed (pi0, pi1[non], gamma1[ovl], gamma2)
- Partial IPW (pi0, pi1, gamma2)
- Fully IPW (pi0, pi1, pi2)
- Oracle (true gamma1, gamma2)

Blip-parameter recovery (centered g-estimation)

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psi2 (gamma2 = psi20 + 0*L2):  
  est = (0.3991, -0.0069)    sd = (0.0540, 0.0442)    true = (0.40, 0.00)  
psi1 (gamma1 = th0 + th1*L1 + th2*A0):  
  est = (0.9007, 0.6992, -0.3959)    sd = (0.0805, 0.0495, 0.1105)  
  true = (0.90, 0.70, -0.40)
```

Takeaways

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- All estimators ~unbiased for the truth (-1.75); blips recovered.
  - Variance ordering proposed < partial IPW < fully IPW, as designed.
  - Bootstrap (which propagates blip-estimation uncertainty) gives coverage near the nominal 0.95 for all estimators.

# Sampling distribution of the target-contrast estimators

Across R = 500 replications (N = 2000)

